Animal Identification Via Camera Scenario

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The **Drone System(DS)'s** end goal is to have a camera that will scan across a set area, which will then send the video to the computer, splice the video into frames, and push each frame into a Convolutional Neural Network (CNN) detector. From there the CNN detector will iterate across the frames, extracting the visuals and 'boxing' the area where an animal might be, then classifying it to the best of its ability with a confidence meter (see Figure 1). Manual classification would also be possible. Then, in our own database, we will store the data for each animal found at this set location/session. Over time, the user will be able to build a profile on the animals found across in the area. Possible API calls would be able to be made as well to see information on the animal (size, migration patterns, endangered status, etc).

A key feature that will be implemented in this scenario would be real time accurate detection of an animal(s). Using Python, we will be using a YOLOv8 model as the CNN detector and libraries such as NumPy and Pandas. As for the hardware, we will be using a programmable camera (ex. Raspberry Pi AI Camera). We only want to test out the model and see if we can have the proper software before getting hands on with a physical drone.

<u>Scenario: Detect an animal species with moderate to high frequency</u>

This is the first major step towards the project with a CNN detector. The purpose of the drone is to identify animals, but there is no stable model at this time. The goal is to get our initial software to do that.

The user starts the program by opening the associated project and running the executable via `python detect_animal.py`. Upon execution, the system will display a start up message giving the program name, the version number, and the initialization status. The program will attempt to connect to the camera that is hooked up to the laptop. Based on the success of the program, the program will either start showing the user a live feed from the camera upon successful connection, or a console error message upon failure.

Upon connection to the camera, the program will send a continuous video back to the computer for the user to see. The program will splice the video into frames and then be sent into the CNN detector. If there is an animal present, the CNN detector will try to put a box around the animal to show that it detects a live animal. In this introductory stage of the project, the user will be able to view the live camera feed with the potential box on any animals identified by the model. This will allow the user to verify that the detection system is working correctly.

To terminate the program, the user can type `Ctrl + C` into the console.

This scenario prioritizes establishing a reliable detection system for the drone rather than end-user functionality. Testing focuses on validating the detection pipeline's accuracy and stability. Future releases will shift toward end-user operation of the deployed drone system.

Figure 1 - User Flow